Remote Sensing Change Detection Method Based on Dynamic Adaptive Focal Loss

Yuqi Xu, Ling Wu, Xiangnan Liu, Chen Chen, Yiman Li, Qian Zhang, Meiling Liu, Tingwei Zhang, and Baowen Yang

***Abstract*— Deep learning (DL) models for change detection (CD) are affected by the changed/unchanged and hard/easy sample imbalance during the training process. Most of the loss functions for solving the sample imbalance problem are a static loss which is difficult to adapt to the variation of data distribution. In this article, we propose a dynamic method termed dynamic adaptive focal loss (DAFL) function. Specifically, we first statistically count the number of changed/unchanged samples in different batches of training data and a dynamic weighting factor is constructed to dynamically and adaptively balance their proportions. Furthermore, a dynamic modulation factor is proposed to suppress the hard/easy sample imbalance. In addition, we employ a CD model based on a progressive scale expansion network (PSENet), which is trained using DAFL for remote sensing images. Experimental results on three CD datasets (CDDs, Sun Yat-sen University (SYSU)-CD, and LEarning VIsion Remote Sensing (LEVIR)-CD) indicate that DAFL outperforms all baseline approaches. Our proposed method achieves the maximum improvement, with an $F1$-score of 0.33%, 0.8%, and 0.94% for sufficient sample size, and 2.15%, 2.61%, and 3.89% for small-sample size, respectively. This advancement is crucial for the application of CD, which provides an alternative method for solving sample imbalance in the condition of varying sample sizes, especially small sample conditions that are common in the real scenario.**

***Index Terms*— Change detection (CD), deep learning (DL), dynamic adaptive focal loss (DAFL), sample imbalance.**

I. Introduction

REMOTE sensing change detection (CD) is the process of discovering and identifying differences in the state of the land surface of the earth by analyzing two or more images of the same scene at different times [1], [2]. It plays a critically important role in the monitoring of forest and vegetation [3], [4], land use and land cover change analysis [5], [6], damage assessment [7], [8], and resource management [9], [10].

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Traditional CD algorithms can be summarized into four categories [11], [12]: 1) image algebra methods [13], [14], [15], [16]; 2) image transformation methods [17], [18]; 3) image classification methods [19], [20]; and 4) image clustering methods [21], [22]. Although the above methods are convenient to implement, simple features are insufficient to satisfy the demand for various and high-accuracy change extraction, so manual feature engineering methods based on machine learning (ML) have been applied to the CD tasks [23], [24], [25]. Nevertheless, ML-based approaches require prior expertise to manually construct and select features, which have a lower generalization performance to different datasets. In recent years, deep learning (DL) has emerged rapidly in the field of artificial intelligence. It offers superior feature representation, model architecture, and performance over the traditional CD methods that require manually extracting features, which enables DL-based methods to effectively perceive the changes in remote sensing images and significantly enhance the accuracy of CD [26], [27]. Convolutional neural network (CNN)-based methods can automatically learn multilevel change information from remote sensing images [28], [29], [30]. For example, introducing residual modules [31] in the network contributes to enhancing model performance, which proposes a novel network ResNet. With its distinctive skip-connection characteristics, it prevents information loss due to deep network hierarchy, making it a hot topic in CD tasks in recent years [32], [33], [34], and frequently serving as the backbone network in CD. Besides, transformers which were originally designed for natural language processing tasks [35] have received extensive attention in the field of computer vision, such as segmentation [36], object recognition [37], classification [38], etc. In contrast to CNNs, transformers have exhibited powerful capabilities in capturing global dependencies to reduce information loss over a long range [39]. Inspired by these works, Chen et al. [40] introduced a bitemporal image transformer into CD tasks.

DL-based CD models demand a substantial volume of training samples, and they specifically demand high-quality training samples. Moreover, the sparse distribution of change areas inherently limits the number of changed samples that can be effectively annotated [41], [42]. As a result, DL models face significant challenges due to two distinct types of sample imbalances during the training process [43]. First, changed/unchanged imbalance, there is an overwhelming surplus of unchanged samples compared to changed samples.

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Excessive unchanged areas will dominate the loss function, which will bias the model toward the unchanged areas, leading to inferior results of CD. Second, there is a considerable surplus of "easy samples" that can be accurately predicted compared to "hard samples" that pose a challenge for accurate prediction. These two imbalances cause the training process to predominantly focus on easy and unchanged samples, resulting in a lack of diversity and useful information during training.

Effectively tackling the aforementioned problem of imbalanced samples is crucial for boosting model performance and enhancing result accuracy. Currently, research primarily addresses the issue of inefficient model training due to sample imbalance from two perspectives: data-level, which involves sample selection to achieve balanced sample sizes, and algorithm-level, which mainly focuses on enhancing the loss function. In data-level approaches, the network is trained by filtering candidate samples through a uniform selection strategy and a random subspace-integrated learning module or by introducing morphological supervision to select equilibrium samples on the boundaries of changed and unchanged regions [22], [44], [45]. The works based on data-level methods mainly focus on addressing the imbalance between changed/unchanged samples without taking into account the imbalance between hard/easy samples. Moreover, the manual workload of a large number of high-quality sample labeling will increase, and it is difficult to represent all types of changes when the number of manually selected samples is limited. Algorithm-level approaches are mainly based on improving or optimizing the loss function that characterizes the weights of the sample categories. The appropriate weights are applied to the loss function by considering the weights of changed and unchanged samples such as the binary cross-entropy (BCE) [46], [47], softmax loss function [44], and triplet loss function [48]. To alleviate the changed and unchanged sample imbalance, a change magnitude-guided loss function was proposed by combining weights to the cross-entropy loss function (WBCE) [49], which accelerates the network training speed and improves the accuracy. A new loss function named focal loss [50] was first proposed in the field of object detection, which balances the changed/unchanged samples by introducing a static weighting factor α. Furthermore, a static modulation factor γ is introduced to moderate the imbalance between hard and easy samples in the training data. Yang et al. [43] proposed a focal weighted loss function that introduces the static modulation factor γ to balance the hard and easy samples into the CD so that the model can be more concerned about hard changed samples during the training process, which enhances the accuracy of the CD results. Although some recent works have considered the imbalance of hard/easy samples and introduced the widely used focal loss into CD tasks, a static modulation factor γ with a constant value is not adapted to the changing data distributions in the face of a dataset with complex scenario that will lead to inconsistent model performance at different training stages, which results in inferior generalization and suboptimal results, limiting the breadth of its application. Therefore, it is necessary to implement dynamic adaptive adjustments of the hyperparameters for the focal loss function to accommodate dynamic changes in the distribution of changed/unchanged samples in different batches as well as hard/easy sample imbalance during the training process.

Given the above issues, we propose a dynamic adaptive focal loss (DAFL) function that acts as a more valid alternative to focal loss for dynamically tackling sample imbalance in CD tasks. To demonstrate the efficiency of the proposed DAFL, we employed a progressive scale expansion network (PSENet) [51] in this work. The main contributions of our work can be summarized as follows.

1) To address the imbalance of changed/unchanged samples and reduce the high demand of DL on the quantity and quality of samples, we introduced the dynamic weighting factor DAFL (α) to dynamically and adaptively adjust the weight based on the varying distribution of changed/unchanged samples in different batches. It allows the model to focus on a smaller number of changed samples and enhances the ability of the model to identify the information of the changed areas.

2) To suppress the impact of hard/easy sample imbalance, we introduced the dynamic modulation factor DAFL (γ) to quantify the expected value of the sample classification difficulty, which is dynamically changed during the training process, to adaptively mine the information of hard samples. Compared with focal loss, our dynamic modulation factor DAFL (γ) is more adaptive to complex datasets, enabling the model to extract more discriminative features.

3) Notably, DAFL achieves excellent results when addressing the sample imbalance in the condition of varying sample sizes.

The rest of this article is organized as follows. Section II introduces our proposed DAFL in detail, while Section III presents the three datasets and experimental settings. Section IV shows our results and performance. Finally, the discussion and conclusion are presented in Sections V and VI.

II. Methodology

In this section, the proposed DAFL is comprehensively introduced. First, an overview of focal loss is presented and explained (see Section II-A). Second, DAFL (α) is proposed to dynamically address the problem that the number of changed pixels is much smaller than the number of unchanged pixels, which is prevalent in current CD datasets (see Section II-B). Third, DAFL (γ) is introduced to dynamically reduce the weights of easy and unchanged samples, enabling the model to focus on hard and changed samples (see Section II-C).

A. Focal Loss

Given the impact of changed/unchanged and hard/easy samples on CD accuracy in the dataset, the focal loss function effectively alleviates the two types of sample imbalance. For convenience of illustration, we unify the probability that the network assigns to the changed or unchanged samples [52] as

$$q_{\text{correct}} = \begin{cases} q, & p = 1 \\ 1 - q, & p = 0 \end{cases} \tag{1}$$

where $q (q \in [0, 1])$ represents the predicted value of the model, and p represents the label with a value of 0 or 1 (0 denotes

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unchanged and 1 denotes changed). The cross-entropy loss function can be simplified by the above formulation as follows:

$$L = -\log q_{\text{correct}}. \tag{2}$$

The main contribution of focal loss is the ability to weigh the loss with a dynamic factor "ω," which balances the hard/easy samples in the network

$$\omega = (1 - q_{\text{correct}})^{\gamma} \tag{3}$$

where $\gamma(\gamma \in [0, +\infty))$ is the modulation factor. For easy and changed samples, increasing γ reduces the impact of easily changed samples on the training process. On the contrary, it can also decrease the impact of easy and unchanged samples on the training process.

In addition, focal loss integrates a weighting factor α which is used to resolve the imbalance between changed and unchanged samples

$$\alpha = \begin{cases} \alpha, & p = 1 \\ 1 - \alpha, & p = 0 \end{cases} \tag{4}$$

where α and $1 - \alpha$ are the weighting factors of changed and unchanged samples, respectively, and $\alpha(\alpha \in [0, 1])$ can be set by the case of different changed and unchanged samples, and finally, the focal loss can be defined as

$$\begin{aligned} L_{\text{focused}} &= -\alpha * \omega * \log q_{\text{correct}} \\ &= -\alpha(1 - q_{\text{correct}})^{\gamma} \log q_{\text{correct}}. \end{aligned} \tag{5}$$

When $\alpha = 0$ and $\gamma = 0$, the focal loss is equivalent to [see (2)].

B. Dynamic Weighting Factor

To adjust α by calculating the ratio of changed/unchanged samples in each batch of data, the α is adjusted. In other words, the dynamic weighting factor DAFL (α) is dynamically weighted based on the difference in the quantity of changed/unchanged samples to ensure that the model dynamically focuses on the fewer changed samples during the training process, preventing them from being overwhelmed by the larger quantity of unchanged samples and achieving a dynamic balance between them. Wang et al. [53] proposed weighted cross-entropy loss (W-CEL) as a dynamic to balance changed/unchanged samples. However, it inevitably introduces weights greater than 1 to the loss function, which poses challenges to effectively converge the loss values during the training process. Therefore, in this article, a new dynamic weighting factor α_c is proposed based on the W-CEL function

$$\alpha_c = \begin{cases} \alpha_P = \dfrac{N_{\text{ratio}}}{P_{\text{ratio}} + N_{\text{ratio}}}, p = 1 \\ \alpha_N = \dfrac{P_{\text{ratio}}}{P_{\text{ratio}} + N_{\text{ratio}}}, p = 0 \end{cases} \tag{6}$$

where $N_{\text{ratio}} = (P + N/N)$ and $P_{\text{ratio}} = (P + N/P)$, P is the total number of "1" in a batch label and N is the total number of "0" in a batch label (i.e., "0" indicates an unchanged sample and "1" indicates a changed sample).

C. Dynamic Modulation Factor

In the early stages of model training, there are sample imbalance problems which results in the bias of model prediction for different classes of samples. To ensure that the dynamic modulation factor DAFL (γ) varies with the training progress, we dynamically adjust it according to the expected value $\hat{q}_{\text{correct}}$ of the true label prediction probabilities from each batch of sample data. In accuracy calculations, regardless of the number of changed/unchanged samples, it is simply obtained by dividing the number of correctly classified samples by the total number of samples. However, when the difference in the number of changed/unchanged samples is large, the accuracy can be dominated by the number of unchanged samples, resulting in a lower assessment of the effectiveness of the detection of changed samples. In this article, a weighting coefficient γ_c is introduced, which affects the model performance by weighing the weights of true positives (TPs) and true negatives (TNs). Specifically, the probability of correct prediction is mapped to a space that is easier to optimize by introducing a logarithmic transformation. It helps to better capture the differences between probabilities during the training process. When facing hard samples with lower expected probability values, γ_c takes on a larger value to grant a larger weight for hard samples. While the expected probability increases, that is, the number of hard samples decreases, the γ_c takes on smaller value, gradually reducing the focus on hard samples. The DAFL (γ) can be defined as

$$\gamma_c = -\log \hat{q}_{\text{correct}} \tag{7}$$

$$\hat{q}_{\text{correct}} = \frac{\text{TP}}{\text{TP} + \text{FP} + r * \text{TN}} \tag{8}$$

where TP, FP, and TN denote the count of TPs, false positives, and TNs, respectively. "r" represents the weighting coefficient for unchanged samples, where r = P/N. When the probability of correct prediction $\hat{q}_{\text{correct}}$ is close to 1, it means that the current model performs better on the CD task, and the value of γ_c will tend to 0; when the probability of correct prediction $\hat{q}_{\text{correct}}$ is close to 0, the value of γ_c will increase, which means that the current model performs poorly on the CD task. It utilizes dynamic γ_c rather than the original fixed value γ. In this case, the model can effectively focus more on the hard samples and thus improving its detection performance. Iteration using exponential smoothing works best if small batches are used for training [52]

$$\hat{q}_{\text{correct}} = 0.95 * \hat{q}^{t-1}_{\text{correct}} + 0.05 * \hat{q}^{t}_{\text{correct}}. \tag{9}$$

The initial value of t is 1 and the final DAFL function is as follows:

$$\begin{aligned} L_{\text{DAFL}} &= -\alpha_c(1 - q_{\text{correct}})^{\gamma_c} \log q_{\text{correct}} \\ &= -\alpha_c(1 - q_{\text{correct}})^{-\log \hat{q}_{\text{correct}}} \log q_{\text{correct}}. \end{aligned} \tag{10}$$

The probability q_{correct} is defined as q when a category is in the changed class and the probability q_{correct} is denoted as $1 - q$ when a category is in the unchanged class. Furthermore, to further mitigate the impact of sample imbalance, we employ a combined loss function, which combines the DAFL function

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Fig. 1. Example samples with the size of 256 × 256 from CDD. First row: Pretemporal images, second row: Posttemporal images, and third row: True labels.

and the dice loss function [54]. The dice loss function can be represented as

$$L_{\text{Dice}} = 1 - \frac{2|X \cap Y|}{|X| + |Y|} \tag{11}$$

where X represents the true label, Y represents the predicted values, and $|X \bigcap Y|$ approximates the dot product and summation of pixel values between the true label and the predicted label.

Therefore, the expression for the loss function that we utilize can be represented as follows:

$$L = L_{\text{DAFL}} + L_{\text{Dice}}. \tag{12}$$

Furthermore, the strategy of using a mixed loss function has proven beneficial in enhancing the accuracy of dual-temporal CD results [55], [56]. Since this article primarily focuses on assessing the effectiveness of the proposed DAFL, all subsequent experiments have employed dice loss. Therefore, the dice loss will not be presented in the subsequent experiments. In summary, DAFL (α) and DAFL (γ) are proposed to balance the effects of changed/unchanged and hard/easy samples to the loss function which can efficiently extract more accurate variation maps.

III. Experiments

In this section, we empirically evaluated our proposed DAFL by comparing it with other commonly used loss functions on three available public CD datasets (CDDs). We presented details on the datasets, backbone network, implementation details and evaluation criteria, and benchmark loss functions.

A. Datasets

To evaluate the proposed DAFL, three public CDDs: CDD [57], Sun Yat-sen University (SYSU-CD) [58], and LEarning VIsion Remote Sensing (LEVIR-CD) [59] were involved in the experiments.

1) CDD: CDD (see Fig. 1) is one of the publicly available assessment datasets in the field of CD. It consists of 11 pairs of dual-temporal multisource remote sensing images from Google Earth (Digital Globe) with resolutions of 0.03–1 m per pixel. Ji et al. [57] processed the raw remote sensing images to generate a total of 16 000 image pairs. The size of each pair of images is 256 × 256 pixels, of which 10 000 pairs of images are used to construct the training set, while the remaining 6000 pairs are used for the test and validation sets, respectively.

TABLE I

Statistics on the Number of Changed Pixels and the Number of Unchanged Pixels in the CDD, SYSU-CDD, and LEVIR-CDD

Dataset		Changed	Unchanged	C/ Uc
CDD	train	83,981,014	571,378,986	0.147
	val	24,676,497	171,800,431	0.144
	test	25,411,239	171,196,761	0.148
	total	134,068,750	914,376,178	0.147
SYSU-CD	train	167,833,309	618,598,691	0.271
	val	56,437,798	205,706,202	0.274
	test	61,820,917	200,323,083	0.309
	total	286,092,024	1,024,627,976	0.279
LEVIR-CD	train	21412971	445203349	0.048
	val	2816268	64292596	0.044
	test	6837404	127380324	0.054
	total	31066643	636876269	0.049

2) SYSU-CDD: SYSU-CDD (see Fig. 2) includes 20 000 pairs of aerial images with 0.5-m resolution obtained from Google Earth and Baidu Maps in 2007 and 2014, and it covers scenes of several cities and areas with image pixels of 256 × 256. Many types of changes in relatively complex scenarios are provided in the dataset such as buildings, vegetation, croplands, roads, and sea. Of these, 12 000 image pairs are used for the training set, and 8000 image pairs are used for the test and validation sets, respectively.

3) LEVIR-CDD: The LEVIR-CDD (see Fig. 3) is a large-scale CDD with 0.5 m spatial resolution. It contains 637 pairs of remote sensing images with the size of 1024 × 1024. The images originated from 20 different areas of several cities in the United States. We split the image into sizes of 256 × 256 with no overlap during processing. We employed a 3:1:1 ratio to split the dataset which assigned 60% of the samples for training, 20% for validation, and 20% for testing.

We counted the volume of changed pixels and unchanged pixels in the CDD, SYSU-CDD, and LEVIR-CDD, and the results are shown in Table I. The ratio of the number of changed pixels to the total number of unchanged pixels were

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Fig. 2. Example samples with the size of 256 × 256 from SYSU-CD. First row: Pretemporal images, second row: Posttemporal images, and third row: True labels.



Fig. 3. Example samples with the size of 256 × 256 from LEVIR-CD. First row: Pretemporal images, second row: Posttemporal images, and third row: True labels.

0.147, 0.279, and 0.049 in the CDD, SYSU-CDD, and LEVIR-CDD which implies that the changed areas are much smaller than the unchanged areas in both datasets.

B. Backbone Network

Due to the rich spectral-spatial-temporal information contained in the high-resolution remote sensing images, more demands are placed on the network in the process of extracting deep-level features. To evaluate the effectiveness of the proposed DAFL, all experiments were conducted on PSENet [51]. It employs a CNN backbone as the feature extractor, which is modified from ResNet-50 [32] by removing the fully connected layer. The residual modules enable the network to finely tune and optimize feature mappings to better extract high-level semantic information and improve the efficiency and performance of feature learning. Then feature pyramid network (FPN) [60] is used to upsample high-level feature maps and fuse them with low-level feature maps. It consists of two components: a bottom-up feature extraction network and a top-down feature fusion network.

C. Implementation Details and Evaluation Metrics

The model presented in this article and all the experiments involved are implemented in the PyTorch 1.8 framework. The AdamW optimizer is adapted with an initial learning rate of 0.0001 and the batch size is configured as 8 to facilitate model convergence. Moreover, warm-up and cosine annealing strategies are applied to optimize the model's performance. All of our experiments are conducted on the NVIDIA GeForce RTX 3070ti.

To quantitatively evaluate the performance of the proposed method, we utilized four evaluation metrics: Precision [see (13)], Recall [see (14)], $F1$-score [see (15)], and intersection over union (IoU) [see (16)]. Precision represents the proportion of samples predicted as changed that are changed. The recall represents the proportion of actual changed samples that are predicted as changed. In the context of CD tasks, higher precision and recall values indicate fewer false positives and false negatives, respectively. The $F1$-score is the harmonic mean of precision and recall, revealing the overall detection performance [61]. IoU represents the degree of overlap between the changed area predicted by the model and the truly changed area. The range of values for IoU and $F1$ is 0 to 1, and the larger their values, the better the CD results

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{13}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{14}$$

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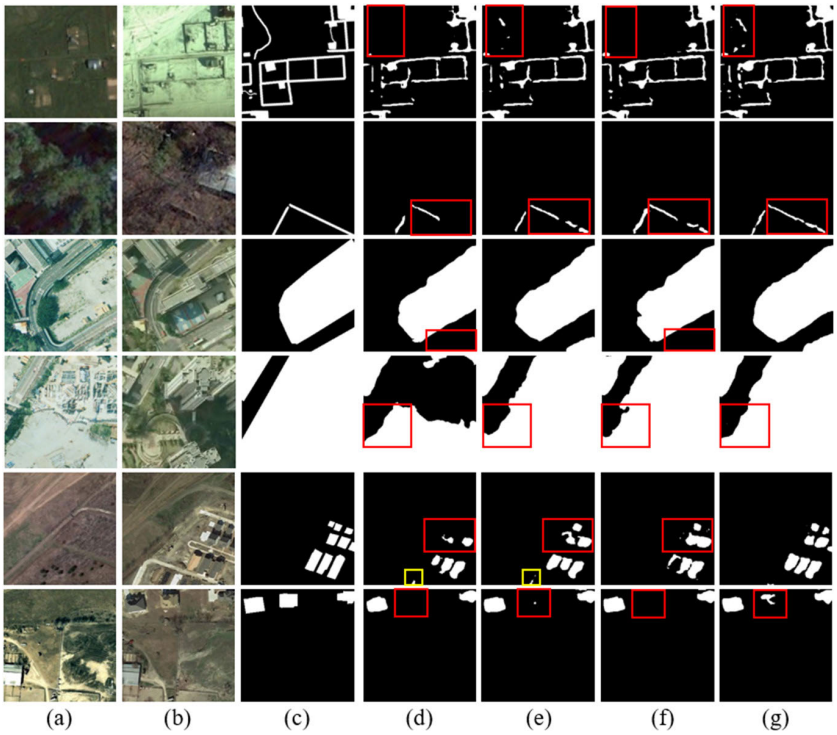


Fig. 4. Visualization results of various loss function methods. The first two rows, the middle two rows, and the last two are example samples of CDD, SYSU-CD, and LEVIR-CDDs, respectively. (a) Pretemporal. (b) Posttemporal. (c) True labels. (d) PSENet with focal loss. (e) PSENet with DAFL (α). (f) PSENet with DAFL (γ). (g) PSENet with DAFL. We plot the omission error in the red box and the commission error in the yellow box.

$$F1-\text{score}=\frac{2\text{Pre}\times\text{Rec}}{\text{Pre}+\text{Rec}} \tag{15}$$

$$\text{IoU}=\frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{16}$$

where TP, FP, TN, and FN represent the count of TPs, false positives, TNs, and false negatives, respectively.

D. Benchmark Loss Functions

We compare the detection results of the proposed DAFL with four loss functions (BCE, WBCE, focal loss, and IoU) in PSENet networks, respectively. A brief description of the above method is as follows.

1) BCE Loss [47]: A method used to measure the similarity between two probability distributions, as defined in [see (2)]. The final loss value is obtained by calculating the cross-entropy between the true and predicted values.

2) WBCE Loss [49]: Similar to [see (6)], it is used to measure the similarity between two probability distributions. It differs from BCE in that it introduces a weighting factor for changed and unchanged samples.

3) Focal Loss [50]: A loss function designed for imbalanced datasets, as defined in [see (5)]. Commonly used to address the issue of class imbalance, especially in object detection tasks, where there is an imbalance between changed and unchanged samples as well as between hard and easy samples.

4) IoU Loss [62]: A loss function utilized to quantify the degree of overlap between the true and predicted values. It measures the accuracy of predictions by calculating the ratio of the intersection to the union between the predicted values and the true areas.

IV. Results

The section aims to provide valuable insights into the effectiveness and potential applications of the proposed DAFL as a regulatory factor. In Section IV-A, ablation experiments were conducted to assess and compare the effectiveness of the proposed regulatory factors, DAFL (α) and DAFL (γ), in a vertical comparison context. In Section IV-B, a horizontal comparison was performed by comparing DAFL with four commonly used loss functions: BCE, WBCE, focal loss, and IoU. In Section IV-C, a detailed comparative experiment was conducted to explore the impact of sample quantity on the performance of DAFL, considering sample sizes ranging from 100 to 8000.

A. Ablation Study on DAFL

To demonstrate the superiority of DAFL (α) and DAFL (γ), we conducted ablation studies on the CDD, SYSU-CD, and LEVIR-CDDs. Among the following experiments, "With DAFL (α)" refers to the focal loss with the dynamic weighting factors, "With DAFL (γ)" denotes the focal loss with the dynamic modulation factor, respectively. All ablation experiments are trained on PSENet and the specific results are shown in Fig. 4 as well as Table II.

From Table II, the proposed DAFL yields more accurate results and helps to enhance the performance of the model on three datasets. When focal loss was employed, the model exhibited the poorest performance. Compared to focal loss, it can be observed that after adding dynamic weighting factor α, the recall, IoU, and $F1$-score of the CDD increased by 0.26%, 0.15%, and 0.27%, respectively. Similarly, the recall,

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TABLE II
LOSS FUNCTION ABLATION RESEARCH ON DAFL

Method	CDD P / R / F1 / IoU(%)	SYSU-CD P / R / F1 / IoU(%)	LEVIR-CD P / R / F1 / IoU(%)
PSENet with Focal Loss	95.83/95.62/95.72/91.80	**81.24**/79.17/80.19/66.93	87.26/86.52/86.89/76.81
PSENet with DAFL (α)	95.87/95.88/95.87/92.07	81.19/79.52/80.34/67.14	86.39/87.84/87.11/77.17
PSENet with DAFL (γ)	95.86/95.69/95.77/91.89	80.58/79.93/80.26/67.02	86.89/87.26/87.07/77.11
PSENet with DAFL	**95.98/95.90/95.94/92.20**	80.62/**80.11/80.36/67.17**	**88.11/87.99/88.05/78.65**

IoU, and $F1$ scores increased by 0.35%, 0.15%, and 0.21% on the SYSU-CDD and 1.32%, 0.22%, and 0.36% on the LEVIR-CDD. Besides, the dynamic modulation factor γ increases the recall, IoU, and $F1$-score of the CDD, SYSU-CDD, and LEVIR-CDD by 0.07%, 0.05%, and 0.09%; 0.76%, 0.07%, and 0.09%; 0.74%, 0.18%, and 0.30%, respectively. This indicates that both variations are more sensitive to hard and changed samples and significantly reduce omission errors. The proposed DAFL achieved the optimal performance in three metrics of recall, IoU, and $F1$-score. Compared to focal loss, the integration of both DAFL (α) and DAFL (γ) leads to an increase of 0.40% and 0.22% in the IoU and $F1$-score on the CDD, and an increase of 0.24% and 0.17% in the SYSU-CDD, respectively. Similarly, on the LEVIR-CDD, there is an increase of 1.84% and 1.16%, respectively. Modifying any hyperparameters in focal loss is able to improve the results performance. In a word, either DAFL (α) or DAFL (γ) is beneficial to enhance the validity of finding all the change regions as much as possible, and DAFL contributes to improving the model performance which confirms the feasibility of integrating DAFL (α) and DAFL (γ).

Conclusions similar to the quantitative analysis results on the three datasets are depicted in Fig. 4. On the CDD, it should be noted that the final results of focal loss are not satisfactory which misses some significant changes [see red boxes in Fig. 4(d)]. In contrast, from Fig. 4(e) and (f), DAFL (α) and DAFL (γ) are capable of detecting subtle changes, indicating that dynamically adjusting hyperparameters during training is beneficial to adapt to different datasets. DAFL combines the advantages of both DAFL (α) and DAFL (γ) to better recognize changed areas with more complete boundaries and still achieve the best visual results, as seen in Fig. 4(g). Since there are various types of variations contained on the SYSU-CDD, the recall and $F1$ scores are not substantially improved. However, the proposed DAFL is much closer to the original labels in the visual results, as shown in Fig. 4. With both the DAFL (α) and DAFL (γ), DAFL performs well on the test set and accurately identifies almost all the changes, as shown in Fig. 4(g). On the LEVIR- CDD, focal loss has more missed detections compared to DAFL (α) and DAFL (γ), as shown in Fig. 4(d)–(f). The proposed DAFL is able to identify changed areas, which further proves its effectiveness in mitigating the imbalance of changed/unchanged and hard/easy samples, especially for more challenging changed areas.

B. Feasibility Analysis of Addressing Sample Imbalance Using the DAFL Function

Quantitative performance analysis of different loss functions based on PSENet on the CDD is shown in Table III. DAFL, while slightly lower in precision than BCE, achieves the best performance regarding recall and $F1$-score. Specifically, it obtains precision, recall, $F1$-score, and IoU of 95.98%, 95.90%, 95.94%, and 92.20%, respectively. Compared with focal loss, the improvements are 0.16%, 0.29%, 0.22%, and 0.40%, respectively. For a more intuitive evaluation, representative experimental results are visualized in Fig. 5. Visually, BCE and WBCE perform reasonably well, but they do not take into account the distribution of both changed/unchanged and hard/easy samples. While some major changes are detected, there are more missed detections and several misclassifications when BCE and WBCE were used [see red and yellow boxes in Fig. 5(d) and (e)]. Focal Loss takes the distribution of both changed/unchanged and hard/easy samples into account simultaneously, making it more effective than the BCE and WBCE. After the proposed method was performed, some edge and minor changed areas can be detected, and the boundaries look more complete, as shown in Fig. 5(h). This is mainly attributed to the real-time and dynamic adjustment of the weights of changed/unchanged and hard/easy samples, which indicates the overall effectiveness of the DAFL for CD in bi-temporal remote sensing images.

The results of quantitative assessment and partial detection of the SYSU-CDD are shown in Table III and Fig. 6. It is evident that the $F1$-score of different loss functions is generally inferior to those of the CDD, primarily due to the diverse ground change types and complex backgrounds. Notably, the best performance of the model was achieved using DAFL, with an improvement of 0.17% in $F1$ score and 0.24% in IoU compared to the focal loss. It validates the effectiveness of the proposed DAFL in addressing the imbalance between changed/unchanged and hard/easy samples. Additionally, Fig. 6 shows the comparison of the visualization results on the SYSU-CDD. In contrast to the CDD, there are noticeable instances of both omission and commission errors in WBCE, BCE, and focal loss, as shown in the red and yellow boxes in Fig. 6(d)–(f). The proposed DAFL function outperforms all comparative methods, yielding relatively complete change regions, as shown in Fig. 6(h).

The results of quantitative assessment and partial detection of the LEVIR-CDD are shown in Table III and Fig. 7. Compared to the suboptimal methods, DAFL shows absolute gains

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TABLE III
EXPERIMENTAL RESULTS ON CDD AND SYSU-CDD. THE VALUES IN BOLD FOLD ARE THE BEST

Method	CDD P / R / F1 / IoU(%)	SYSU-CD P / R / F1 / IoU(%)	LEVIR-CD P / R / F1 / IoU(%)
PSENet with BCE	**96.06**/95.22/95.64/91.64	81.09/79.36/80.22/67.01	86.21/87.58/86.89/76.82
PSENet with WBCE	95.83/95.30/95.56/91.51	80.96/77.82/79.36/65.78	87.75/86.49/87.11/77.17
PSENet with Focal Loss	95.82/95.61/95.72/91.80	81.24/79.17/80.19/66.93	87.26/86.52/86.89/76.81
PSENet with IOU	95.88/95.63/95.75/91.85	**81.92**/78.41/80.13/66.84	87.46/86.75/87.10/77.15
PSENet with DAFL	95.98/**95.90/95.94/92.20**	80.62/**80.11/80.36/67.17**	**88.11/87.99/88.05/78.65**

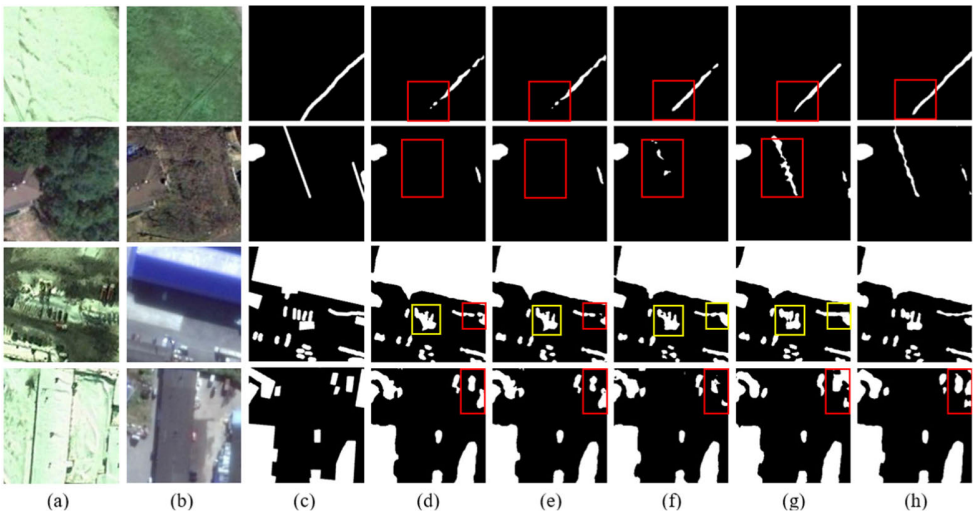


Fig. 5. Visualization results of various loss function methods on CDD. (a) Pretemporal. (b) Posttemporal. (c) True labels. (d) PSENet with BCE. (e) PSENet with WBCE. (f) PSENet with focal loss. (g) PSENet with IoU. (h) PSENet with DAFL. We plot the omission error in the red box and the commission error in the yellow box.

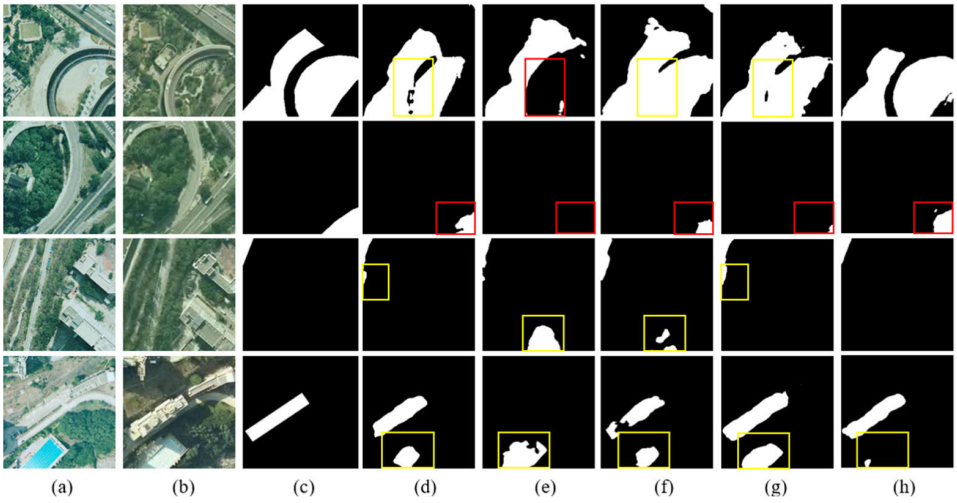


Fig. 6. Visualization results of various loss function methods on SYSU-CDD. (a) Pretemporal. (b) Posttemporal. (c) True labels. (d) PSENet with BCE. (e) PSENet with WBCE. (f) PSENet with focal loss. (g) PSENet with IoU. (h) PSENet with DAFL. We plot the omission error in the red box and the commission error in the yellow box.

of 0.36%, 1.50%, 0.94%, and 1.48% on the four-evaluation metrics, respectively. Other loss functions miss some subtle changed areas and produce false detections in certain areas, as shown in red boxes in Fig. 7(f). This is because the focal loss is a static loss function that cannot dynamically adjust hyperparameter values during the training process, leading to the model incorrectly predicting some changed areas as unchanged. Our proposed method achieves excellent visual results, as shown in Fig. 7(h).

C. Feasibility Analysis of Addressing Sample Imbalance in the Condition of Varying Sample Size Using the DAFL Function

1) Experiment on CDD: We evaluated the feasibility of DAFL in resolving sample imbalance in the condition of infinitely approaching small-sample size scenarios. Quantitative metrics based on precision, recall, $F1$-score, and IoU along with detection results on the CDD are shown in Table IV and Fig. 8. DAFL achieves the best $F1$-score and IoU at

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TABLE IV
EXPERIMENTAL RESULTS ON TWO DIFFERENT DATASETS WITH DIFFERENT NUMBER OF TRAINING SAMPLES. THE VALUES IN BOLD FOLD ARE THE BEST

Number of training samples	Method	CDD P / R / F1 / IoU(%)	SYSU-CD P / R / F1 / IoU(%)	LEVIR-CD P / R /F1 / IoU(%)
100	BCE	76.22/60.42/67.41/50.83	51.32/60.75/55.64/38.54	50.33/44.35/47.15/30.85
	WBCE	76.05/60.28/67.25/50.66	52.68/59.92/56.07/38.96	45.85/47.54/46.68/30.44
	Focal Loss	77.00/60.25/67.60/51.06	48.96/61.89/54.67/37.62	52.15/34.23/41.33/26.05
	IoU	73.61/58.17/64.98/48.13	**56.24**/52.90/54.52/37.48	49.70/46.58/48.09/31.66
	DAFL	**77.01/60.83/67.97/51.48**	50.80/**64.87/56.98/39.84**	**56.84/47.89/51.98/35.12**
200	BCE	78.36/69.87/73.87/58.57	43.49/49.27/46.20/30.04	48.75/46.90/47.80/31.41
	WBCE	79.50/69.46/74.14/58.91	57.04/50.30/53.46/36.48	50.69/45.39/47.89/31.49
	Focal Loss	78.99/71.07/74.82/59.77	53.76/50.85/52.26/35.38	**52.87**/44.15/48.11/31.68
	IoU	78.23/71.37/74.64/59.54	50.39/44.48/47.25/30.93	47.63/46.02/46.81/30.56
	DAFL	**81.17/73.19/76.97/62.56**	**58.04/50.94/54.26/37.23**	51.25/**47.67/49.39/32.80**
300	BCE	77.64/72.90/75.20/60.25	48.20/46.11/47.13/30.83	56.81/60.33/58.51/41.36
	WBCE	79.10/71.75/75.25/60.32	51.71/47.29/49.40/32.80	59.56/60.17/59.87/42.72
	Focal Loss	**81.45**/69.87/75.22/60.28	51.12/41.99/46.11/29.96	56.56/61.32/58.85/41.69
	IoU	75.63/73.44/74.52/59.39	49.56/47.36/48.44/31.96	54.73/61.85/58.07/40.92
	DAFL	79.67/**74.26/76.87/62.43**	**52.92/51.13/52.01/35.14**	**61.38/65.45/63.35/46.36**
400	BCE	80.71/80.30/80.51/67.37	61.35/60.41/60.87/43.75	59.55/62.22/60.85/43.73
	WBCE	81.18/80.18/80.68/67.62	63.96/58.61/61.17/44.06	66.46/65.41/65.93/49.18
	Focal Loss	79.75/80.36/80.05/66.74	62.00/59.11/60.52/43.39	**68.86**/55.93/61.72/44.64
	IoU	**83.16**/77.21/80.08/66.77	**65.90**/57.25/61.27/44.17	64.11/67.02/65.54/48.74
	DAFL	79.65/**82.12/80.87/67.88**	64.54/**62.19/63.34/46.35**	64.60/**70.67/67.50/50.94**
500	BCE	83.28/80.40/81.81/69.22	**63.59**/67.99/65.71/48.94	66.85/63.41/65.09/48.24
	WBCE	83.57/80.44/81.98/69.46	58.75/73.91/65.46/48.66	70.22/66.37/68.24/51.79
	Focal Loss	**85.73**/77.15/81.21/68.37	61.99/70.02/65.76/48.99	68.28/64.29/66.23/49.51
	IoU	80.63/79.88/80.26/67.02	57.75/74.63/65.11/48.27	69.01/66.64/67.80/51.29
	DAFL	83.63/**80.73/82.15/69.71**	61.92/**75.88/68.19/51.73**	**70.60/67.37/68.94/52.61**
600	BCE	**85.87**/79.18/82.39/70.06	73.25/70.71/71.96/56.20	71.42/59.99/65.21/48.38
	WBCE	84.02/81.56/82.77/70.61	72.79/72.99/72.89/57.34	71.96/64.15/67.83/51.32
	Focal Loss	83.93/81.37/82.63/70.41	**76.07**/71.65/73.80/58.43	75.20/62.80/68.44/52.03
	IoU	84.64/77.65/81.00/68.06	68.38/73.62/70.90/54.92	71.64/64.61/67.95/51.45
	DAFL	85.27/**81.68/83.44/71.58**	74.40/**74.16/74.28/59.08**	**76.53/65.01/70.30/54.20**
700	BCE	86.44/81.45/83.87/72.23	64.47/79.15/71.06/55.11	72.84/67.47/70.05/53.91
	WBCE	86.32/80.19/83.14/71.15	64.45/78.76/70.89/54.91	75.84/66.51/70.87/54.88
	Focal Loss	83.14/81.70/82.42/70.09	**68.10**/75.57/71.64/55.81	**77.95**/62.78/69.54/53.31
	IoU	81.63/79.29/80.44/67.29	65.31/75.95/70.23/54.12	73.45/64.53/68.70/52.32
	DAFL	**86.92/82.11/84.45/73.08**	67.72/**79.20/73.01/57.49**	77.33/**67.97/72.35/56.67**
800	BCE	84.98/82.46/83.70/71.97	70.30/67.23/68.73/52.36	71.60/71.95/71.77/55.98
	WBCE	85.07/82.53/83.78/72.09	69.73/71.64/70.67/54.65	75.55/71.61/73.52/58.13
	Focal Loss	**87.19**/81.66/84.33/72.91	58.03/69.41/63.21/46.21	74.34/69.50/71.83/56.05
	IoU	86.10/81.79/83.89/72.25	60.98/70.35/65.33/48.51	69.89/72.74/71.29/55.39
	DAFL	86.43/**82.75/84.55/73.23**	**70.32/72.72/71.50/55.64**	**75.59/72.90/74.22/59.01**
900	BCE	87.84/83.13/85.42/74.55	72.52/72.04/72.28/56.59	73.04/75.40/74.20/58.98
	WBCE	87.37/83.07/85.17/74.16	72.17/73.97/73.06/57.55	74.82/75.50/75.16/60.20
	Focal Loss	87.58/82.69/85.07/74.02	74.55/68.41/71.35/55.46	73.83/74.98/74.40/59.24
	IoU	**88.52**/82.19/85.24/74.28	69.48/73.83/71.59/55.75	73.56/76.14/74.83/59.78
	DAFL	88.12/**83.16/85.57/74.78**	**75.11/74.19/74.65/59.55**	**76.25/76.45/76.35/61.75**
1000	BCE	88.04/84.47/86.22/75.78	70.64/72.81/71.71/55.89	74.02/77.21/75.59/60.75
	WBCE	88.85/83.67/86.18/75.72	71.30/72.37/71.83/56.04	78.59/76.46/77.51/63.28
	Focal Loss	89.02/82.99/85.90/75.28	70.28/73.50/71.86/56.08	81.50/77.20/79.29/65.69
	IoU	88.31/84.07/86.14/75.65	**75.86**/66.54/70.89/54.91	77.40/76.65/77.02/62.63
	DAFL	**89.51/84.48/86.92/76.87**	72.42/**74.06/73.23/57.77**	**82.11/77.67/79.83/66.42**
2000	BCE	92.02/88.04/89.99/81.79	**75.27**/68.84/71.91/56.14	85.22/80.32/82.70/70.50
	WBCE	92.22/87.94/90.03/81.87	71.20/72.81/71.99/56.42	86.57/81.59/84.01/72.42
	Focal Loss	91.80/87.35/89.52/81.03	74.03/68.80/71.32/55.42	83.62/81.44/82.51/70.23

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TABLE IV
(Continued.) EXPERIMENTAL RESULTS ON TWO DIFFERENT DATASETS WITH DIFFERENT NUMBER OF TRAINING SAMPLES. THE VALUES IN BOLD FOLD ARE THE BEST

	IoU	92.12/87.82/89.92/81.68	74.41/69.18/71.70/55.88	85.11/81.17/83.09/71.07
	DAFL	**92.37/88.43/90.36/82.41**	72.02/**73.57/72.79/57.22**	**87.10/81.63/84.28/72.82**
4000	BCE	94.08/91.83/92.94/86.82	80.45/70.14/74.94/59.93	83.46/86.71/85.05/73.99
	WBCE	94.11/91.87/92.97/86.87	82.10/69.96/75.55/60.71	83.58/85.35/84.46/73.10
	Focal Loss	**94.43**/91.40/92.89/86.72	80.82/69.97/75.00/60.00	**85.61**/84.76/85.18/74.19
	IoU	93.94/91.83/92.87/86.69	81.15/69.51/74.88/59.85	83.15/86.73/84.90/73.76
	DAFL	94.12/**91.93/93.02/86.94**	**82.17/70.55/75.92/61.18**	84.19/**86.79/85.47/74.63**
	BCE	94.59/93.86/94.23/89.08	76.33/75.19/75.76/60.97	86.21/87.58/86.89/76.82
	WBCE	94.41/93.65/94.03/88.73	78.88/74.57/76.66/62.16	87.75/86.49/87.11/77.17
6000	Focal Loss	94.72/93.13/93.92/88.53	81.24/72.82/76.80/62.34	87.26/86.52/86.89/76.81
	IoU	94.26/93.84/94.05/88.76	**82.83**/71.10/76.52/61.97	87.46/86.75/87.10/77.15
	DAFL	**94.73/93.89/94.31/89.23**	78.26/**76.16/77.20/62.86**	**88.11/87.99/88.05/78.65**
	BCE	95.48/94.88/95.18/90.80	81.59/75.53/78.44/64.53	
	WBCE	95.36/94.90/95.13/90.71	80.84/75.83/78.26/64.28	
8000	Focal Loss	95.20/94.50/94.85/90.20	81.75/75.06/78.26/64.29	
	IoU	95.36/94.24/94.80/90.11	81.55/74.87/78.07/64.03	
	DAFL	**95.56/94.97/95.26/90.95**	**81.91/75.94/78.81/65.03**	

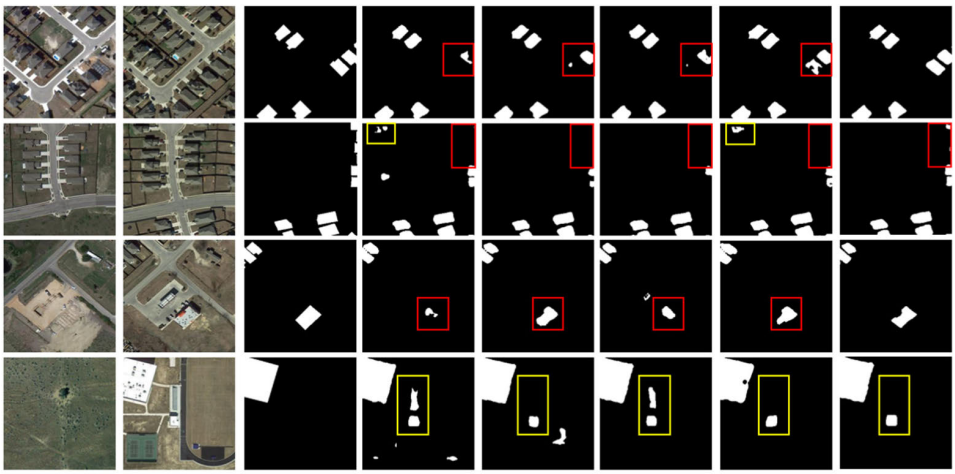


Fig. 7. Visualization results of various loss function methods on LEVIR-CDD. (a) Pretemporal. (b) Posttemporal. (c) True labels. (d) PSENet with BCE. (e) PSENet with WBCE. (f) PSENet with focal loss. (g) PSENet with IoU. (h) PSENet with DAFL. We plot the omission error in the red box and the commission error in the yellow box.

different quantities of training samples. As the number of training samples continues to decline, there is an inevitable overall drop in both $F1$-score and IoU. However, DAFL still outperforms the comparison loss functions. The results demonstrate the superiority of our DAFL in addressing sample imbalance in the condition of varying sample sizes. Simultaneously, we plot the differences between DAFL and the optimal values among four loss functions in precision, recall, $F1$-score, and IoU on the CDD in Fig. 9. The accuracy differences significantly fluctuate in the scenario of small sample size. However, with the increase in the sample size, the differences gradually diminish and eventually stabilize. Notably, the accuracy improvement of DAFL is more apparent in the case of small samples, with the most substantial improvement observed at a sample size of 200. From Fig. 8, it can be seen that the results generated by BCE, WBCE, focal loss, and IoU are not satisfactory by visualizing the experimental results with a training sample size of 200 pairs. The main changed areas are not detected and some unchanged samples are predicted as changed, which means there are partial commission errors, as indicated in the red and yellow boxes in Fig. 8(d)–(g). In this case, it is still feasible to identify most of the changed areas using DAFL, as shown in Fig. 8(h).

2) Experiment on SYSU-CDD: Quantitative evaluations and partial detection results on the SYSU-CDD are presented in Table IV and Fig. 10. DAFL achieves the optimal detection results for almost all evaluation metrics in different sample sizes, and this improvement is remarkable which are identical to the CDD. We plot the differences between DAFL and the optimal values among four loss functions in precision, recall, $F1$-score, and IoU on the SYSU-CDD, as shown in Fig. 11. It is evident that DAFL further exhibits a more pronounced improvement when there are fewer training samples. This illustrates that DAFL can be regarded as an effective method

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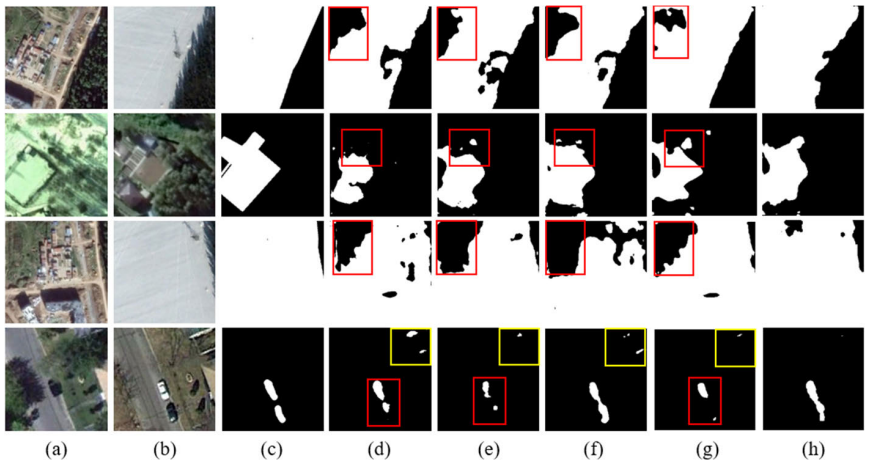


Fig. 8. Visualization results of various loss function methods on CDD with 200 training samples. (a) Pretemporal. (b) Posttemporal. (c) True labels. (d) PSENet with BCE. (e) PSENet with WBCE. (f) PSENet with focal loss. (g) PSENet with IoU. (h) PSENet with DAFL. We plot the omission error in the red box and the commission error in the yellow box.

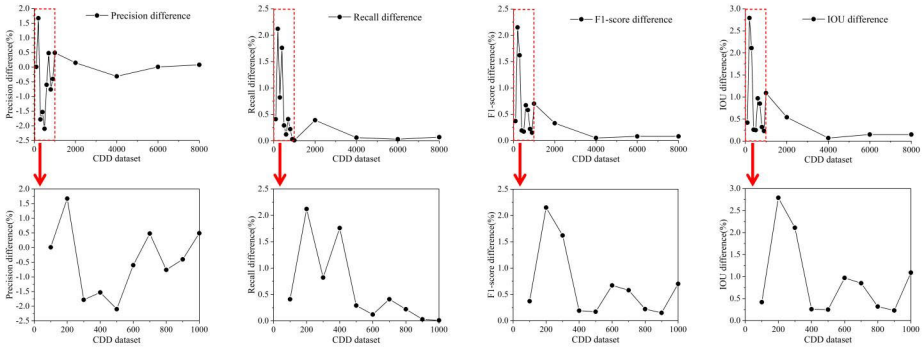


Fig. 9. Subtracted results between DAFL and the most superior loss function on CDDs.

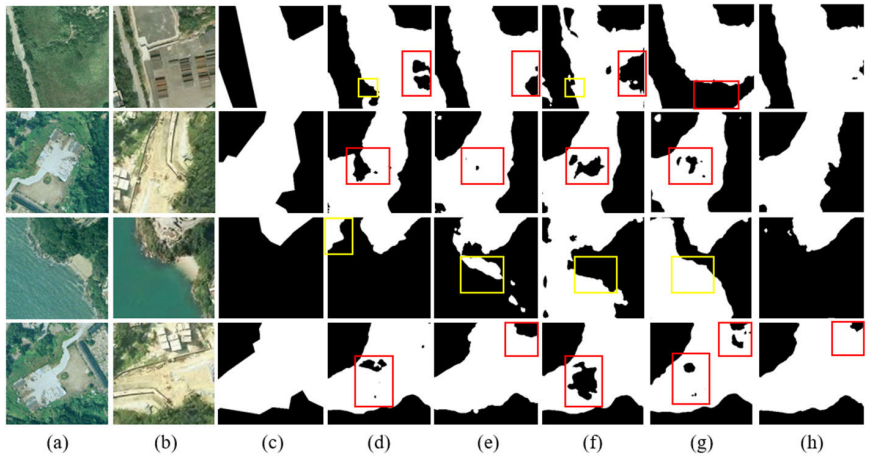


Fig. 10. Visualization results of various loss function methods on SYSU-CDD with 300 training samples. (a) Pretemporal. (b) Posttemporal. (c) True labels. (d) PSENet with BCE. (e) PSENet with WBCE. (f) PSENet with focal loss. (g) PSENet with IoU. (h) PSENet with DAFL. We plot the omission error in the red box and the commission error in the yellow box.

for sample imbalance in the condition of varying sample sizes. Unlike the CDD, DAFL achieves the greatest improvement among all the experiments when the number of training sample sizes is 300 pairs. Due to the more complex scenarios of the SYSU- CDD, a significant amount of changed information is lost in the detection results of BCE, WBCE, focal loss, and IoU, as shown in the red boxes in Fig. 10(d)–(g). It is acceptable because some changes are hard to recognize even with human eyes. The proposed method in this article provides more details and is closer to real-world conditions. Therefore, it obtains more complete and accurate changed areas than other methods.

3) Experiment on LEVIR-CDD: The results of the comparison experiments of different loss functions in LEVIR-CD are listed in Table IV. When referring to the IoU metric, DAFL achieves the best IoU at different quantities of training samples. We plot the differences between DAFL and the optimal values among four loss functions in precision, recall, $F1$-score, and IoU on the LEVIR-CDD, as shown in Fig. 12. When the number of training samples size is 100 pairs, our

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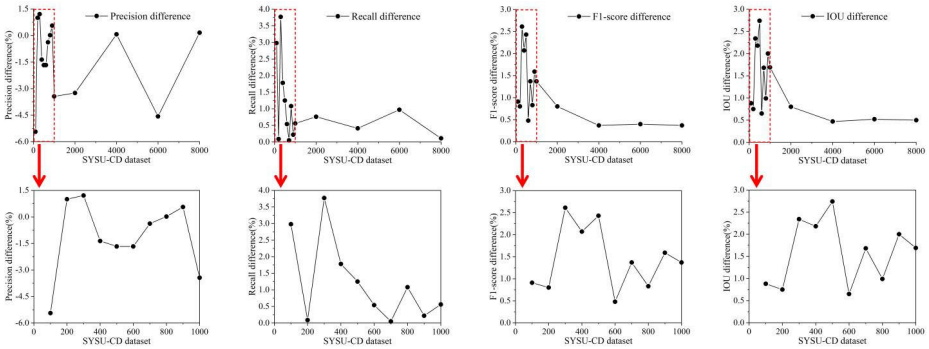


Fig. 11. Subtracted results between DAFL and the most superior loss function on SYSU-CDDs.

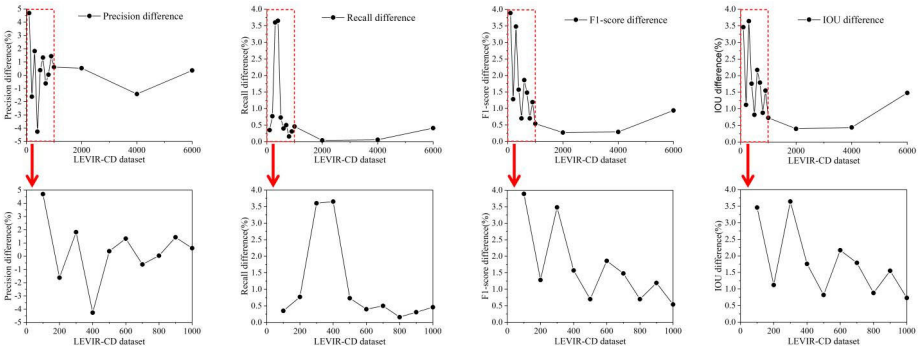


Fig. 12. Subtracted results between DAFL and the most superior loss function on LEVIR-CDDs.

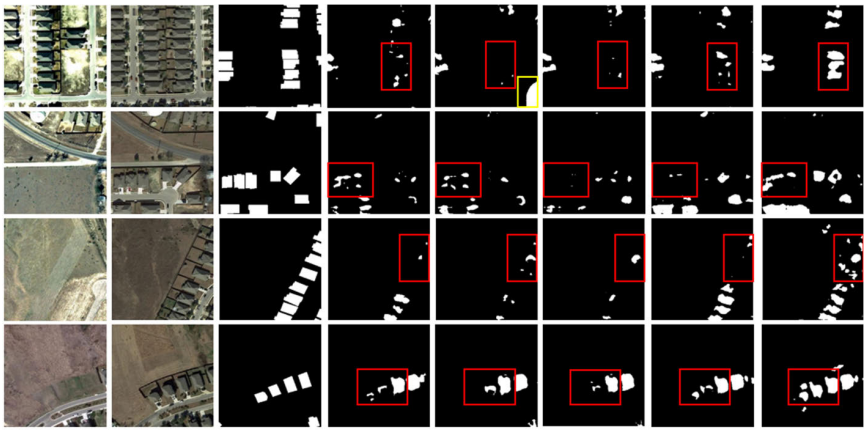


Fig. 13. Visualization results of various loss function methods on LEVIR-CDD with 100 training samples. (a) Pretemporal. (b) Posttemporal. (c) True labels. (d) PSENet with BCE. (e) PSENet with WBCE. (f) PSENet with focal loss. (g) PSENet with IoU. (h) PSENet with DAFL. We plot the omission error in the red box and the commission error in the yellow box.

DAFL achieves optimal performance. It corresponds to 4.69%, 0.35%, 3.89%, and 3.46% growth in precision, recall, $F1$-score, and IoU, respectively. As shown in Fig. 13(h), the proposed DAFL is capable of inferring the changed areas relatively completely, indicating that it is applicable to detect changes in the case of a small sample size.

V. Discussion

In practical engineering applications, our priority is to enhance recall rather than precision. It is due to the fact that high recall means low omission errors and the most of changed areas are detected. The commission errors (corresponding to precision) can be easily corrected manually; however, the correction of omission errors (undetected changed areas) needs more manual workload. Therefore, it is vital to ensure that the model can identify all the changed areas as much as possible and that the proposed method improves the recall and possesses the applied potential. Although our proposed DAFL function achieved a good CD effect, it still has some limitations. First, although DAFL has the highest accuracy among all comparison methods when the sample size is small, the accuracy is relatively low compared to the results from a large sample size. Currently, semi-supervised CD methods can make full use of large amounts of unannotated data for model training under the guidance of less annotated data, which alleviates the requirement for annotation [63], [64]. For instance, to make full use of the unannotated data, Zhang et al. [63] proposed a framework that consists of a supervised branch for annotated data and an unsupervised branch for unannotated data. The unsupervised branch uses

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weakly augmented unannotated remote sensing image pairs to generate pseudo labels to supervise two strongly enhanced counterparts.

Furthermore, although DAFL can alleviate the sample imbalance in the condition of varying sample size to a certain extent using PSENet in this article, it is not guaranteed that it is completely suitable for other CD models. Therefore, in the subsequent work, we might continue to apply DAFL to three commonly utilized CD methods such as fully convolutional early fusion (FC-EF) [65], fully convolutional siamese concatenation (FC-Siam-Conc) [65], and fully convolutional siamese difference (FC-Siam-Di) [65] to verify its effectiveness.

Currently, most DL-based CD tasks are conducted on the datasets and are not applied to real-world scenarios to validate the effectiveness of the methods. CD in more complex real-world scenarios is more valuable compared to the ideal dataset. The proposed DAFL loss function in this article is mainly focused on the case of sample imbalance and small sample size which is more aligned with real-world situations. The validity of the proposed method was confirmed in the ideal dataset, and therefore the validity of it in more complex real-world scenarios needs to be proved in the future. In addition, it is also crucial to take into account the transferability of the proposed method in real-world scenarios. There might be various issues with the application of DAFL to the open world that require further research.

VI. Conclusion

In this article, we proposed a novel loss function DAFL as an effective solution for resolving changed/unchanged and hard/easy sample imbalances. It accomplishes this by solving both changed/unchanged and hard/easy sample imbalance in CD tasks using the loss function beforehand. In DAFL, we employed DAFL (α) to dynamically weight changed/unchanged samples, which helps to protect the underrepresented changed class and penalize the overrepresented unchanged one. To mitigate the effect of hard/easy samples on network training, we proposed a dynamic modulation factor DAFL (γ), whose uniqueness lies in preventing premature convergence caused by very low loss values in the late stages of training to ensure the model reaches its optimum accuracy. The effectiveness of the proposed DAFL (α) and DAFL (γ) was confirmed in ablation experiments. Furthermore, PSENet can extract multiscale and deep-level features which enable the network to accurately identify and analyze complex land surface changes. Compared to four commonly used loss functions, PSENet with DAFL achieved superior performance on the CDD, SYSU-CD, and LEVIR-CDDs. In summary, DAFL has the capability to dynamically adapt to different training progresses. It not only enhances CD accuracy under sample imbalanced conditions but also improves the model's ability to recognize changes when oriented toward the sample imbalance in the condition of varying sample size.

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